

RAFFLES INSTITUTION
2012 YEAR 6 PRELIMINARY EXAMINATION

Higher 1



CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Paper 2 Structured Questions

14 September 2012
2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Answer paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces provided at the top of this page.
Write in dark blue or black pen in the spaces provided.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions in the answer booklet.

Section B

Answer **two** questions on separate answer paper.

A Data Booklet is provided. Do not write anything on it.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use		
Paper 1	/ 30	
Paper 2 (circle the questions you have answered)	A1	/ 9
	A2	/ 5
	A3	/ 15
	A4	/ 11
	B5	/ 20
	B6	/ 20
	B7	/ 20
Total	/ 100	

Section A (40 marks)Answer **all** the questions in this section in the spaces provided.

- 1 (a) Complete the following table.

Particle	$^{31}_{15}\text{P}^{3-}$	$^{59}_{27}\text{Co}^{2+}$
Number of neutrons		
Number of electrons		

[4]

- (b) When potassium manganate(VII) is dissolved in very concentrated aqueous potassium hydroxide, the manganate(VII) ion is reduced to form green MnO_4^{2-} . When this strongly alkaline solution is diluted with water, a brown solid is precipitated and the solution turns purple.

Identify the brown precipitate and the purple solution.

Brown precipitate:

Purple solution:

[2]

- (c) With the aid of relevant half-equations, construct a balanced equation for the reaction that occurs when the green solution containing MnO_4^{2-} ions is diluted.

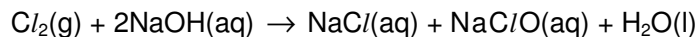
[3]**[Total: 9]**

- 2 Chlorine and fluorine react exothermically to form a strongly oxidising interhalogen compound, ClF_3 .
When gaseous ClF_3 is added to water, a vigorous reaction occurs, giving three gases as the only products. One of the gases turns moist blue litmus paper red and another gas bleaches it. The third gas relights a glowing splint.

- (a) Construct a balanced equation, including state symbols, for the reaction.

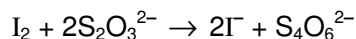
.....**[1]**

- (b) Sodium chlorate(I), NaClO, can be prepared in the laboratory by bubbling chlorine gas into cold aqueous NaOH.



The concentration of NaClO prepared can be determined by reacting it with excess acidified potassium iodide to form chloride ions and iodine as the products. The iodine liberated is then titrated against sodium thiosulfate solution, Na₂S₂O₃(aq).

It was found that 25.0 cm³ of the NaClO solution liberated iodine that required 50 cm³ of 0.100 mol dm⁻³ of sodium thiosulfate solution according to the equation:



- (i) State the type of reaction undergone when chlorine gas is bubbled into cold aqueous NaOH.

.....

- (ii) Write a balanced equation for the reaction between chlorate(I) and iodide ions in acidic medium.

.....

- (iii) Calculate the concentration of the sodium chlorate(I) solution.

[4]
[Total: 5]

- 3 (a) Butene gas, C_4H_8 , and butane gas, C_4H_{10} , are commonly used as fuels. The standard enthalpy change of combustion of butene is $-2715 \text{ kJ mol}^{-1}$.

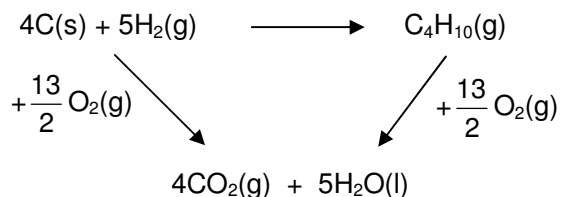
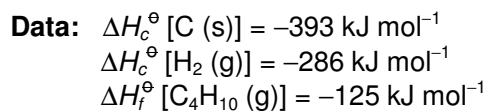
(i) What is the minimum mass of butene required to bring 100 g of water at 0°C to 100°C ? Assume that the process is 80 % efficient and that the specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

(ii) Assuming no heat loss to the surroundings, briefly explain why the mass of butene required to bring 100 g of ice at 0°C to boiling is likely to be higher than that calculated in (a)(i).

.....
.....

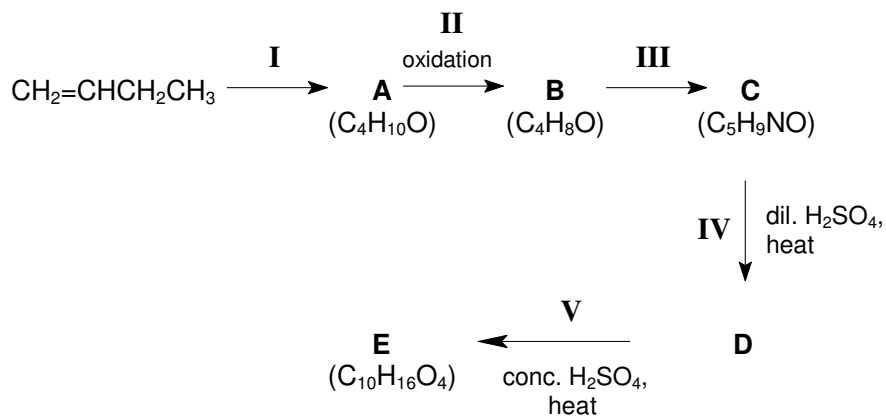
(iii) The heat released when 1 g of a substance is combusted is known as its fuel value (in kJ g^{-1}). Calculate the fuel value of butene.

- (iv) Using the following information and energy cycle, calculate the standard enthalpy change of combustion of butane.



[6]

- (b) But-1-ene can be converted to a cyclic compound **E** via the following series of reactions.



- (i) Both compounds **A** and **B** produce a yellow precipitate on warming separately with alkaline aqueous iodine. State the reagents and conditions for steps **I** and **III**.

Step **I**:

Step **III**:

- (ii) In the boxes below, draw the structural formulae of compounds **B**, **C**, **D** and **E**.

B	C
D	E

An isomer of but-1-ene can exhibit geometrical isomerism.

- (iii) Draw and label the structure of **both** geometrical isomers.

- (iv) Briefly explain which geometrical isomer will have a higher boiling point.

[9]
[Total: 15]

- 4 Although having the same number of carbon atoms, ethanoic acid and 2-chloroethanoic acid have significantly different pK_a values at 298 K.

compound	formula	pK_a
ethanoic acid	CH_3COOH	4.79
2-chloroethanoic acid	$CH_2ClCOOH$	2.86

- (a) (i) Suggest why 2-chloroethanoic acid has a smaller pK_a value than ethanoic acid.

- (ii) Suggest a reagent that can be added to ethanoic acid to produce an acidic buffer solution.

.....

- (iii) With the aid of equations, briefly explain how the acidic buffer solution from your answer to (a)(ii) controls pH.

[5]

- (b) The experimental results of a kinetic study of the reaction between substance **A** and substance **B** are given in the table below.

experiment number	[A] / mol dm ⁻³	[B] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.100	0.150	6 x 10 ⁻³
2	0.150	0.150	9 x 10 ⁻³
3	0.200	0.200	1.6 x 10 ⁻²

- (i) Use the data to deduce the order of reaction with respect to both **A** and **B**.

order of reaction with respect to **A**

order of reaction with respect to **B**

- (ii) Hence calculate the value of the rate constant, k , for the reaction and give its units.

[6]
[Total: 11]

Section B (40 marks)

Answer any **two** questions from this section on separate answer paper.

- 5 (a) The average oxidation number of the two sulfur atoms in sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, is +2.

When solid sulfur containing the radioactive isotope ^{35}S is boiled with sodium sulfite, $\text{Na}_2^{32}\text{SO}_3$, a radioactive sample of sodium thiosulfate is produced. Adding $\text{HCl}(\text{aq})$ to this sample causes all the ^{35}S to precipitate as sulfur. The resulting solution contains non-radioactive sulfite ions.

- (i) Draw the dot-and-cross diagram of the sulfite ion.
- (ii) State the shape and bond angle of the sulfite ion.
- (iii) Use your answer to (a)(i) and the reactions described to suggest a structure for the thiosulfate ion produced. On your structure, label any radioactive atom.

[4]

- (b) (i) Sketch the graphs illustrating the trends in atomic radius from sodium to chlorine and explain the general trend.
- (ii) The melting points of a few Period 3 oxides are given below.

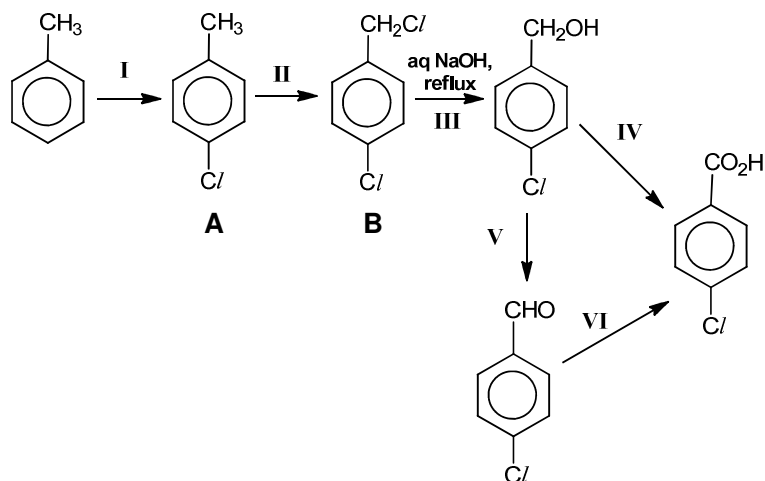
compound	formula	m.p. / °C
sodium oxide	Na_2O	1275
aluminium oxide	Al_2O_3	2072
sulfur dioxide	SO_2	-73

Outline the acid-base nature of the oxides, giving equations where appropriate.

- (iii) Describe the reactions, if any, of magnesium chloride and silicon tetrachloride with water. Write equations where appropriate and predict the pH of the resulting solution.

[9]

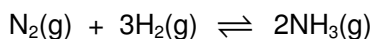
- (c) The following shows two reaction schemes for the synthesis of 4-chlorobenzoic acid.



- (i) Suggest suitable reagents and conditions for steps I, II, IV and V of the reaction scheme above.
- (ii) Suggest how compounds **A** and **B** can be distinguished by a simple chemical test.
- (iii) When compound **B** was treated with another reagent, a quaternary ammonium salt, $(C_6H_4CH_2)_4N^+Cl^-$ was produced. Name the reagent and give the conditions required to bring about this reaction.

[7]
[Total: 20]

- 6 In the Haber process, nitrogen and hydrogen are combined to produce ammonia, a reaction that is represented by the following equation.



- (a) Using relevant bond energy data from the *Data Booklet*, determine the enthalpy change for the above reaction.

[2]

A catalyst is also used in the process. The catalyst used is usually iron that is promoted by other compounds, which improve the efficiency of the iron catalyst.

- (b) (i) Explain how this is an example of *heterogeneous catalysis*.
(ii) Using your answer to (a), sketch a labelled energy profile diagram of the *uncatalysed* and *catalysed* reactions.

[4]

- (c) With the aid of a Maxwell-Boltzmann distribution curve, explain why the rate doubles when there is an increase of 10 °C.

[3]

- (d) The following shows the first six ionisation energies of an element **X** in Period 3.

942 1733 3387 4511 16247 19940

- (i) State which group element **X** belongs to. Briefly explain your answer.
(ii) Write down the electronic configuration of the valence shell.

Element **X** forms a compound with carbon. The compound has a melting point of 2730 °C and has a hardness which approaches that of diamond.

- (iii) Explain, in terms of bonding and structure, why the compound formed between **X** and carbon has such a high melting point.
(iv) Describe one possible use for the compound formed between **X** and carbon.

[6]

- (e) Beryllium forms simple dihalides of formula BeX_2 with all the halogens.

- (i) Describe and explain the shape of the BeF_2 molecule.
(ii) BeF_2 and ammonia, NH_3 , react in a 1:2 ratio to give a white crystalline solid. Draw a diagram to show the bonding within a molecule of this solid, explaining the type of bonding involved.

[5]

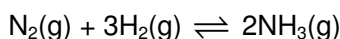
[Total: 20]

- 7 (a) 0.10 mol of an organic compound **A**, $C_xH_yO_z$, gives 0.40 mol of carbon dioxide and 0.40 mol of water vapour upon complete combustion. 8.8 g of **A**, when vaporised, was found to occupy 2.4 dm^3 at room temperature and pressure.

Compound **A** does **not** react with aqueous sodium hydroxide. However, it forms a yellow precipitate when warmed with alkaline aqueous iodine. **A** reacts with SOCl_2 and gives off gaseous products. A silver mirror is seen when **A** is warmed with Tollens' reagent. Compound **B**, an isomer of **A**, is neutral to litmus. On treatment with hot aqueous sodium hydroxide, **B** yields **C**, $\text{C}_2\text{H}_3\text{O}_2\text{Na}$, and **D**, $\text{C}_2\text{H}_6\text{O}$.

- (i) Determine the molecular formula of **A**.
 (ii) Deduce, with reasoning, the structures of **A**, **B**, **C** and **D**.
 Chemical equations are not required. [10]

- (b) Ammonia is manufactured by the Haber process.



The table below shows the number of moles of ammonia in equilibrium mixtures at various temperatures and pressure. In each case, 1.5 mol of N_2 and 3.0 mol of H_2 were mixed in a 3 dm^3 vessel initially.

Pressure / atm	No. of moles of ammonia at equilibrium		
	at 300 °C	at 400 °C	at 500 °C
1	0.55	0.34	0.13
100	1.00	0.60	0.30
200	1.63	1.05	0.71

- (i) Using Le Chatelier's Principle, state and explain the effect on the equilibrium position for the above reaction when a higher pressure is used at constant temperature.
 (ii) Using the information in the table, deduce and explain whether the forward reaction is an endothermic or exothermic reaction. Hence, using Le Chatelier's Principle, state and explain the effect on the equilibrium position for the above reaction when a higher temperature is used at constant pressure.
 (iii) By considering the rate of reaction and the yield, suggest the optimum conditions of temperature and pressure for the commercial production of ammonia via the Haber process. Briefly explain your choice for the optimum conditions.
 (iv) Calculate the value of K_c at 200 atm and 300 °C, stating its units. [10]

[Total: 20]

END OF PAPER